

Jack Lutz

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Education

University of California, Los Angeles

Expected Graduation June 2027

B.S. Statistics and Data Science, Minor in Mathematics GPA: 3.5, Major GPA: 3.8

Los Angeles, CA

- **Relevant Coursework:** Probability Theory, Mathematical Statistics, Experimental Design, Abstract Linear Algebra, Differential Equations, Vector Calculus, Data Analysis and Regression, Game Theory

Experience

New York Life Insurance Company

Summer 2026

Incoming AI/Machine Learning Engineering Intern

Philadelphia, PA

Researcher, UCLA Department of Mathematics

Spring 2026

Researcher

Los Angeles, CA

- Researching statistical arbitrage in cryptocurrency markets: constructing normalized spread series across liquid trading pairs, testing for mean reversion via cointegration analysis and walk-forward evaluation, with survivorship-bias-free dynamic universe construction across 100+ spot pairs.

Lutz Consulting Group

Nov 2024 – Present

Founder

Las Vegas, NV

- Analyzed 138 months of transaction data; built Gradient Boosting forecasting model with engineered lag, rolling, and seasonal features, achieving 11.5% MAPE, 58% improvement over naive baseline on 24-month holdout validation, uncovering \$550K+ in dead stock for liquidation.

New York Life Insurance Company

June 2025 – Aug 2025

Data Analyst Intern

New York, NY

- Built RAG-enabled underwriting chatbot prototype using company underwriting manual as knowledge base, demonstrating feasibility of LLM-powered policy guidance. Finalist in company-wide AI Innovation Competition.
- Designed controlled experiments measuring LLM sycophancy: varied prompt authority framing across 3 models, measured response drift toward user-implied conclusions, and produced the team's first empirical model behavior finding. Results drove prompt engineering policy for production deployment.

Projects

Quantitative Research Backtester & Platform | Python, AWS, TypeScript, C++, FastAPI, Vite

2025 – Present

- Built full-stack quant research platform: strategy IDE, real-time charting, and MCP AI integration layer on top of co-developed C++ backtesting engine with custom DSL and per-fold walk-forward validation; data pipeline spanning equity, crypto, macro, and fundamental data across 13,000+ instruments.
- Secured seven-figure AUM allocation from institutional partner (Mazer Group); live pilot generating positive alpha against S&P 500 benchmark.
- Architected autonomous research pipeline with model-tiered routing: cheap, fast models for pre-screening and early termination of weak strategies, larger models for structured reasoning and nuanced validation, reducing wasted compute on low-signal ideas while preserving rigor across 27 MCP-orchestrated tools.
- Designed context hierarchy balancing token cost against reasoning depth: L0 stateless tool calls, L1 session memory, L2 full research context; with prompt caching and programmatic tool calling, cut per-run cost from ~\$10 to ~\$0.50.

UCLA Dining Recommendation App | Python, Go, FastAPI, AWS

2025 – Present

- Co-founded UCLA dining startup; built full backend with Go scraper ingesting 38,000+ menu items daily and CP-SAT constraint solver generating macro-optimized meal plans in 52ms average solve time.

Skills

Languages: Python, R, SQL, TypeScript, JavaScript, Go, C++, C

ML/AI: PyTorch, Scikit-Learn, Statsmodels, RAG, multi-agent orchestration, MCP, prompt caching, LangChain, Hugging Face, FAISS, pandas, NumPy, Claude Code

Infrastructure: FastAPI, PostgreSQL, SQLite, Redis, Docker, AWS, Git, React, Linux

Leadership: Eagle Scout | President, Bruin AI Research Club (Led DataFest team, individual problem-solving award 2023) | Lambda Chi Alpha Executive Board